SIPHOSETHU THOBELANI DLAMINI

Computer Systems Engineering Student — WIL Intern, 4IR Research Lab (Agentic AI)

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SUMMARY

Final-year Computer Systems Engineering student at Tshwane University of Technology (TUT), currently completing Work-Integrated Learning (WIL) at the 4IR Research Lab, Department of Agentic AI, and building a career in artificial intelligence, generative AI, and agentic AI. Maintained a 72% academic average with 11 distinctions, and received the 2024 TUT ICT Faculty Top Academic Achiever Award. Practical experience building AI agent workflows with Python and LangGraph, REST APIs with FastAPI, full-stack web applications with React and Docker, and computer vision systems with OpenCV. Certified in Python, Generative AI, AI Agents, and API development by Analytics Vidhya. Eager to keep learning from experienced developers and to contribute directly to team projects.

TECHNICAL SKILLS

- **AI & Agent Systems:** Generative AI, Python AI Agents, LangGraph, Tool Calling, RAG (Retrieval-Augmented Generation), OpenAI Agents SDK, Prompt Design
- **Programming Languages:** Python, JavaScript, C / C++, SQL (PostgreSQL, MySQL, SQLite), HTML5, CSS3
- **Backend & Web:** FastAPI, Flask, Node.js, Express, REST APIs, React, Vite
- **Databases:** PostgreSQL, pgvector, ChromaDB, MySQL, SQLite, Cloud Firestore
- **Developer Tools:** Docker, Docker Compose, Git, GitHub, Linux, Postman, VS Code
- **Computer Vision:** OpenCV, Haar Cascades face detection, LBPH face recognition
- **Hardware & Embedded IoT:** ESP32, ATmega328P microcontrollers, Sensors, Proteus simulation

PROJECTS

UniAgent — University Student AI Platform | *Personal Project*

Technologies: Python, FastAPI, LangGraph, PostgreSQL, pgvector, Docker, React

- Built an AI platform with 7 helper agents to assist university students with academic advice, career guidance, and course questions.
- Set up document search (RAG) using PostgreSQL and pgvector so the system answers questions accurately from university handbooks without guessing.
- Added human-confirmation steps before the system carries out important actions like submitting forms.
- Created fast backend APIs using FastAPI and tested endpoints to keep response times quick.

Smart Exam Venue Access System (SRSEVAR) | *Academic Project*

Technologies: Python, Flask, OpenCV, SQLite, HTML, CSS

- Created a face recognition check-in system using Python, Flask, and OpenCV to identify students entering exam rooms in real time.
- Trained face detection and LBPH recognition models to match student faces against stored student profile photos.
- Automatically recorded attendance with date and time in SQLite to replace slow paper-based sign-in lists.

MediCare HMS — Hospital Management System | *Full-Stack Project*

Technologies: React, Vite, Firebase (Auth, Firestore, Hosting), Node.js, Express, MySQL, Docker Compose

- Built a hospital management web application with 14 sections for patients, appointments, pharmacy stock, lab results, and billing.
- Added secure login and user roles so staff members (doctors, nurses, pharmacists) see only the tools and records they need.
- Packaged the frontend, backend, and MySQL database into Docker containers for easy installation and testing.
- Moved the system to Firebase (Authentication, Firestore with security rules, Hosting) so it runs live for free, and added patient self sign-up with a patient portal.
- **Live:** medicare-hms-18.web.app

Smart House & IoT Automation | *Hardware & Embedded Project*

Technologies: Embedded C/C++, ESP32, Sensors, Proteus

- Programmed an ESP32 microcontroller in C/C++ to read temperature, light, and motion sensors and send live readings over Wi-Fi.

EDUCATION

Tshwane University of Technology (TUT) — Pretoria, South Africa | Final Year (2026)
National Diploma: Computer Systems Engineering

- **Academic Record:** 72% Average • 11 Distinctions Earned
- **WIL Status:** Completed all theory modules; now completing WIL at the 4IR Research Lab, Department of Agentic AI.
- **Key Subjects:** Database Systems, Microcontrollers, Digital Systems, Network Systems, Engineering Mathematics.

AWARDS & ACHIEVEMENTS

- **Top Academic Achiever Award (2024):** Awarded by the Tshwane University of Technology ICT Faculty for outstanding academic results.
- **11 Academic Distinctions:** Earned 11 distinctions across Computer Systems Engineering subjects.

EXPERIENCE & LEADERSHIP

WIL Intern — *4IR Research Lab, Department of Agentic AI (2026 – Present)*

- Member 2 (Retrieval & Knowledge Base) on Agentic QA-RAG, a five-member research project building an agentic question-answering system for SAHPRA Module 3.2.S regulatory documents.
- Own the Evidence/RERA agent, the S.4 Control agent and the ChromaDB knowledge base that the other agents retrieve from.

Laboratory Tutor Assistant — *TUT (2024 – Present)*

- Helped students during computer and electronics practicals, assisted with microcontroller wiring, and showed students how to debug code errors.

Academic Tutor — *Student Academic Support Office (SASO), TUT (2024 – Present)*

- Provided one-on-one and small group tutoring in programming logic, helping students understand concepts and improve their marks.

Residence Mentor — *Student Affairs & Residence Life, TUT (2024 – Present)*

- Mentored new students, helping them transition into university life, organize their study schedules, and resolve day-to-day residence issues.

CERTIFICATIONS (ANALYTICS VIDHYA)

- **Claude Code: The Coding Assistant** | Analytics Vidhya (Issued: Aug 2, 2026) — Verify Certificate
- **Introduction to Python** | Analytics Vidhya (Issued: Aug 4, 2026) — Verify Certificate
- **Generative AI - A Way of Life** | Analytics Vidhya (Issued: Aug 5, 2026) — Verify Certificate
- **FastAPI for AI Engineers: The Complete Guide to Building Scalable AI APIs** | Analytics Vidhya (Issued: Aug 13, 2026) — Verify Certificate
- **Claude 4.5: Smarter, Faster & More Human AI** | Analytics Vidhya (Issued: Aug 13, 2026) — Verify Certificate
- **Demystifying OpenAI Agents SDK** | Analytics Vidhya (Issued: Aug 14, 2026) — Verify Certificate

REFERENCES

- **Molebogeng Mashilo** — SASO Coordinator, TUT • Mashilom1@tut.ac.za • 076 666 3171
- **Tadiwanashe Songore** — SASO Supervisor, TUT • SongoreTB@tut.ac.za • 062 962 2755
- **Ipeleng Machele** — Lecturer: Computer Systems, TUT • macheleil@tut.ac.za • 061 255 8713